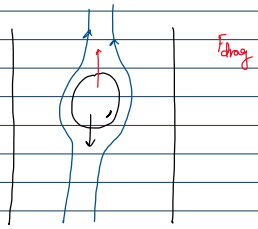
Fundamental Lab (FL): Experiment 1 → Viscosity of fluids: Stoke's law

Drag force exerted on a spherical object in a viscous fluid



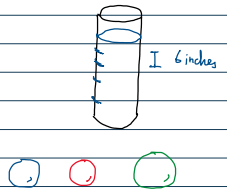
Empirically defined for fully laminar flow conditions ($Re < 0.1$)

$$F_D = 3\pi \mu D v$$

- D: diameter of sphere
- v: velocity of sphere
- μ: Fluid viscosity → Dependent on temperature!

Lab:

+ 3 columns filled w/ liquid (of varying viscosities)
+ Spherical particles of different materials, diameter
+ Electronic balance, tweezers, calipers, stop watch



Determine estimates for viscosity

Viscosity and Stokes Law



Logistics:

+ No Make up sessions! [1 hr 50 min] → 5 min early, Lockers provided, Proper PPE! (You will be sent home otherwise!)
  - PPE: + Safety glasses + Long pants + Closed toe shoes
+ Lab groups → established via Brightspace
+ Complete the trainings → material uploaded on Brightspace

Assignment:

+ Lab Report (due 2 weeks after your Lab → uploaded on brightspace/gradescope)
  - ↳ Introduction
  - ↳ Theory
  - ↳ Methods and Materials
  - ↳ Results and Discussion (w/ graphs, error calculation)
  - ↳ Conclusion & Next steps
  - ↳ Appendix (Derivations, error propagation)

↳ References

+ Group Work Expectation:

↳ Meet up before the lab to discuss responsibilities

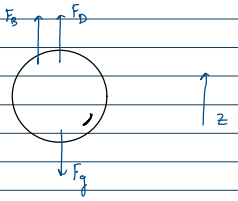
↳ Keep each other accountable

↳ Divide up work to utilize each other's strengths (help out on deficiencies)

Pre-lab assignment

→ How can we go from measurable quantities to determine μ?

$F_D = 3\pi\mu D v$



Terminal velocity → $\Delta F = 0$

$F_B + F_D = F_g$

F_B: Archimedes → Upward force related to the volume of water displaced (equal to the volume of submerged object)

F_D: Stoke's Law

F_g: $mg = (\rho_{ball} V) g$

→ Write viscosity as a function of ρ_{liq} & ρ_{ball}?

$F_B = \rho_{liq} V g$ (m_{liq})

$F_B + F_D = F_g$

$[(\rho_l V) g] + [3\mu\pi D v] = (\rho_{ball} V) g$

$\mu = \dfrac{(\rho_l V g - \rho_{ball} V g)}{3\pi D v}$

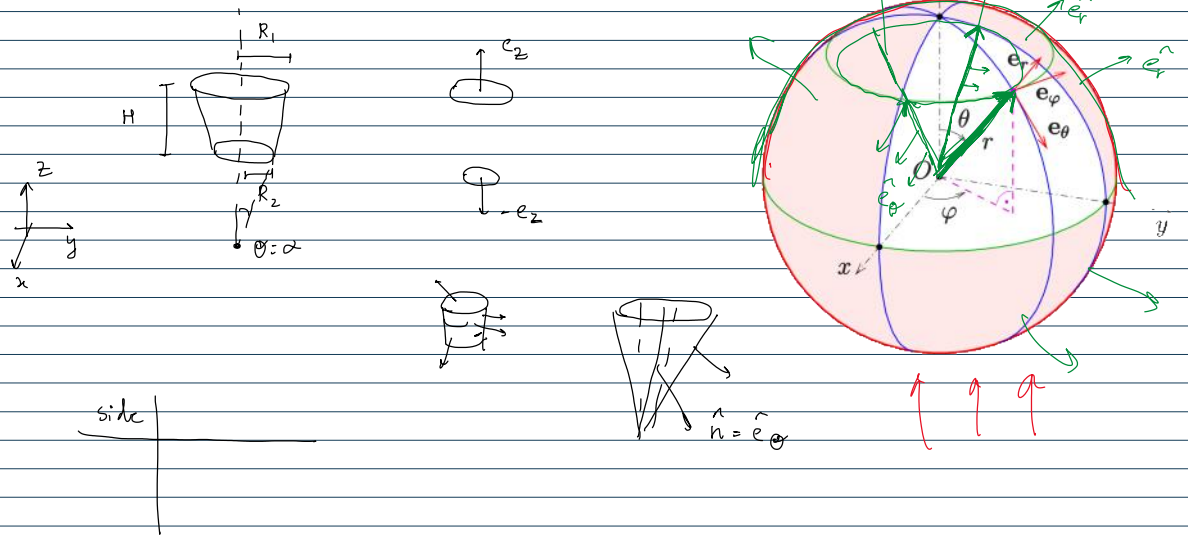
$$\mu = \frac{(\rho_l - \rho_b) V g}{3\pi D v}$$

measure, measure, vary

Interesting analysis: μ varying w/ v
μ varying w/ D
v varying w/ D

PROBLEM: 3.39 For a particular flow problem, the control volume is chosen to be a truncated cone of height H, bottom widest radius R_1, and top smaller radius R_2. The cone is truncated a distance l from the tip, and the cone angle is $\theta \equiv \alpha$, where θ is the coordinate variable for a spherical coordinate system with origin at the cone tip. For each enclosing control surface, write the unit vectors that are normal to each control surface. For a uniform flow $v = -U\hat{e}_z$ down the axis of the control volume, what is the mass flow rate through each control surface? The fluid has constant density ρ, and the flow first passes through the bottom of the control volume.



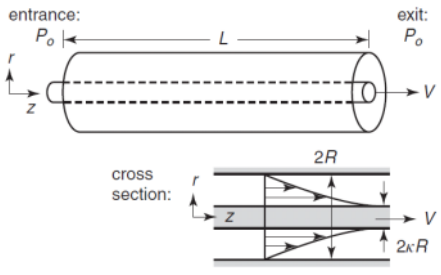


$$\hat{\boldsymbol{r}} = (\sin\theta\cos\phi)\hat{\boldsymbol{x}} + (\sin\theta\sin\phi)\hat{\boldsymbol{y}} + (\cos\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\theta}} = (\cos\theta\cos\phi)\hat{\boldsymbol{x}} + (\cos\theta\sin\phi)\hat{\boldsymbol{y}} - (\sin\theta)\hat{\boldsymbol{z}}$$
$$\hat{\boldsymbol{\phi}} = -(\sin\phi)\hat{\boldsymbol{x}} + (\cos\phi)\hat{\boldsymbol{y}}.$$

PROBLEM: 3.23 For the velocity-profile function $v_z(r)$ given here, what is the average value $\langle v_z \rangle$ of the function between $r = 5$ and $r = 10$? The function is written in cylindrical coordinates. The units on velocity are m/s and the units of r are m.

$$v_z(r) = 8\ln\left(\frac{r}{3}\right)$$

PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?



PROBLEM: 3.32 For the cylindrical coordinate system shown in Figure 3.39 for the axial flow in a wire-coating operation, what is a unit vector in the direction of gravity? What is the component of gravity in the flow direction?

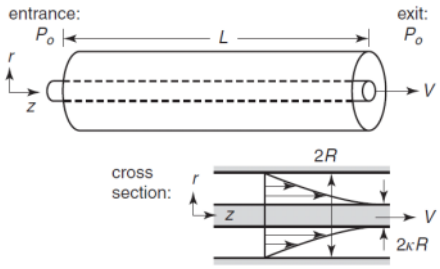


Figure 3.39 Axial annular flow that occurs in wire coating (Problem 32).